

Think preprints are unreliable? Analysis of 70,000 studies might change your mind

Central conclusions rarely change when biomedical preprints go on to be published in journals.

By Mohana Basu • *Nature* (News)

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The central conclusions of biomedical preprints rarely change after peer review in a journal, according to a study posted on the preprint server bioRxiv this month. The research also found that studies that appeared first as preprints are retracted at roughly half the rate of papers that did not appear online before being in a peer-reviewed journal. The authors say the findings suggest that preprints are a reliable source of information, although some scientists say the finding should be interpreted more cautiously.

Posting preprints is common practice in science these days, but Ruslan Rust, a neuroscientist at the University of Southern California in Los Angeles, says that he often hears fellow scientists say that they are unreliable. In his experience, peer review does not typically lead to major changes in the content of a study. Rust wanted to see if this held true across different fields of biomedical research.

Using a large language model (LLM), Rust and his colleague extracted the main scientific conclusion from the abstracts of 72,644 biomedical manuscripts that were first uploaded to bioRxiv between 2018 and 2025. The model then assessed how much the abstracts differed from their eventual peer-reviewed versions. The study, which has not been peer reviewed, reports that 39.9% of main conclusions were unchanged between the preprint and journal-published abstract; another 50% underwent only minor revisions. Slightly more than 10% went through major changes.

When conclusions did change, they were more likely to become more cautious than more confident after peer review, the study found. Around 8.4% of main findings adopted more cautious language after peer review, whereas 4.2% used more confident wording.

The extent of revision also varied between disciplines. Major changes occurred in only 7.2% of bioinformatics papers, compared with 17.5% of microbiology studies.

The authors also found that the frequency of major revisions declined over time, reducing from 17% among papers posted in 2019 to 5.7% in those posted in 2024.

Julian Sienkiewicz, who studies artificial-intelligence tools and data exploration at the Warsaw University of Technology, says the decrease in major revisions over time could indicate that peer reviewers are overloaded and might not be reading papers thoroughly.

Rust suggests that the decrease reflects a change in how people use preprints. In earlier years after bioRxiv was launched, particularly during the COVID-19 pandemic, scientists were under pressure to post their findings in a short time. This meant many papers had to undergo major revisions before publication, he added. In the past few years, some manuscripts might have already incorporated reviewer feedback into the first preprint version that was posted online, he adds.

Fewer retractions

Rust and his colleague also found that papers that first appeared as preprints were retracted at a rate of 8.1 per 10,000 papers, compared with 18.7 per 10,000 comparable papers that had never been posted as preprints.

The authors caution that the comparison is observational, based on relatively few retractions and does not prove that posting a paper as a preprint reduces the likelihood of a retraction.

Reacting to the findings on the professional-networking platform LinkedIn, some researchers noted that preprints are subject to strong selection bias, depending on who posts them and which studies are posted.

Rust agrees that differences between the types of author who choose to post preprints and those who do not could contribute to the pattern his team found. “I think those who post preprints are open about sharing data,” says Rust. “In general, there is evidence that sharing raw data leads to better and more reproducible science,” he adds.

The study only included preprints that were eventually published in a journal and so did not assess the veracity of unpublished preprints. Rust acknowledges this is a limitation of the analysis but adds that previous estimates suggest that roughly two-thirds of preprints posted before 2017 on bioRxiv are eventually published in peer-reviewed journals. “For most scientists, reputation is important, and they would not upload something they would not be comfortable with publishing in a journal later.”

Sienkiewicz also says that LLMs do not always accurately match preprints to their corresponding peer-reviewed versions. Sienkiewicz was part of the team that created PreprintToPaper, a data set that connects bioRxiv preprints to published papers. He adds that in a large data set of thousands of papers even a small margin of error would leave out many preprints from the analysis and possibly skew the result.

Rust says it is important that scientists read papers and evaluate the work themselves, rather than judging the quality of the paper on the basis of whether it was published as a preprint or not. He also suggests using extra caution when assessing preprints to help make decisions in clinical settings.

References

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